

Домашнее задание №8.

Вариант 99.

$$A = 3,2, B = 0,037$$

1. Формат Ф1

$$A = (3.2)_{10} = (3,333333)_{16} = (0,3333333)_{16} \cdot 16^1$$

0	1	0	0	0	0	0	1	0	0	1	1	0	0	1	1
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$$B = (0.037)_{10} = (0,0978D5)_{16} = (0,978D5)_{16} \cdot 16^{-1}$$

0	0	1	1	1	1	1	1	1	0	0	1	1	0	0	0
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$$X_C = X_A - X_B + d$$

$$d + P_C = \frac{P_A + d - P_B - d}{P_C} + d$$

$$X_C = 1 - (-1) + 64 = 66$$

$$P_C = 2$$

№ шага	Действие	Делимое	Частное
0	M_A $[-M_B]_{\text{доп}}$ R_0	0 0 0 1 1 0 0 1 1 1 0 1 1 0 1 0 0 0 1 1 0 0 1 1 0 1 1	0 0
1	$\leftarrow R_0$ $[M_B]_{\text{пр}}$ R_1	1 0 0 1 1 0 1 1 0 0 1 0 0 1 1 0 0 0 1 1 1 0 0 1 1 1 0	0 0
2	$\leftarrow R_1$ $[M_B]_{\text{пр}}$ R_2	1 1 0 0 1 1 1 0 0 0 1 0 0 1 1 0 0 0 0 0 0 1 1 0 1 0 0	0 1
3	$\leftarrow R_2$ $[-M_B]_{\text{доп}}$ R_3	0 0 1 1 0 1 0 0 0 1 0 1 1 0 1 0 0 0 1 1 1 0 1 0 0 0 0	0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
4	$\leftarrow R_3$ $[M_B]_{\text{пр}}$ R_4	1 1 0 1 0 0 0 0 0 0 1 0 0 1 1 0 0 0 0 0 0 1 1 1 0 0 0	0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 1 0 1
5	$\leftarrow R_4$ $[-M_B]_{\text{доп}}$ R_5	0 0 1 1 1 0 0 0 0 1 0 1 1 0 1 0 0 0 1 1 1 0 1 1 0 0 0	0 0 0 0 1 0 1 0 0 0 0 0 1 0 1 0 0 0 0 0 1 0 1 0

6	$\leftarrow R_5$	1 1 0 1 1 0 0 0 0	0 0 0 1 0 1 0 0
	$[M_B]_{np}$	0 1 0 0 1 1 0 0 0	
	R_6	0 0 1 0 0 1 0 0 0	0 0 0 1 0 1 0 1
7	$\leftarrow R_6$	0 1 0 0 1 0 0 0 0	0 0 1 0 1 0 1 0
	$[-M_B]_{доп}$	1 0 1 1 0 1 0 0 0	
	R_7	1 1 1 1 1 1 0 0 0	0 0 1 0 1 0 1 0
8	$\leftarrow R_7$	1 1 1 1 1 0 0 0 0	0 1 0 1 0 1 0 0
	$[M_B]_{np}$	0 1 0 0 1 1 0 0 0	
	R_8	0 1 0 0 0 1 0 0 0	0 1 0 1 0 1 0 1

$$C = (0,55)_{16} \cdot 16^2 = 85.$$

Определим абсолютную и относительную погрешности результата:

$$\Delta C = 86,48648649 - 85 = 1,48648649$$

$$\delta C = \left| \frac{1,48648649}{86,48648649} \right| \cdot 100\% = 1,71875\%$$

2. Формат Ф2

$$A = (3.2)_{10} = (3,333333)_{16} = (0,110011001100110011)_2 \cdot 2^2$$

0	1	0	0	0	0	0	1	0	1	0	0	1	1	0	1
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$$B = (0.037)_{10} = (0,0978D5)_{16} = (0,1001011111001)_2 \cdot 2^{-4}$$

0	0	1	1	1	1	1	0	0	0	0	1	1	0	0	0
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$$X_C = X_A - X_B + d$$

$$d + P_C = \frac{P_A + d - P_B - d}{P_C} + d$$

$$X_C = 2 - (-4) + 128 = 134$$

$$P_C = 6$$

№ шага	Действие	Делимое	Частное
0	M_A	0 1 1 0 0 1 1 0 1	0 0 0 0 0 0 0 0
	$[-M_B]_{доп}$	1 0 1 1 0 1 0 0 0	
	R_0	0 0 0 1 1 0 1 0 1	0 0 0 0 0 0 0 1
1	$\leftarrow R_0$	0 0 1 1 0 1 0 1 0	0 0 0 0 0 0 1 0
	$[-M_B]_{доп}$	1 0 1 1 0 1 0 0 0	
	R_1	1 1 1 0 1 0 0 1 0	0 0 0 0 0 0 1 0

2	$\leftarrow R_1$	1 1 0 1 0 0 1 0 0	0 0 0 0 0 1 0 0
	$[M_B]_{np}$	0 1 0 0 1 1 0 0 0	
	R_2	0 0 0 1 1 1 1 0 0	0 0 0 0 0 1 0 1
3	$\leftarrow R_2$	0 0 1 1 1 1 0 0 0	0 0 0 0 1 0 1 0
	$[-M_B]_{доп}$	1 0 1 1 0 1 0 0 0	
	R_3	1 1 1 1 0 0 0 0 0	0 0 0 0 1 0 1 0
4	$\leftarrow R_3$	1 1 1 0 0 0 0 0 0	0 0 0 1 0 1 0 0
	$[M_B]_{np}$	0 1 0 0 1 1 0 0 0	
	R_4	0 0 1 0 1 1 0 0 0	0 0 0 1 0 1 0 1
5	$\leftarrow R_4$	0 1 0 1 1 0 0 0 0	0 0 1 0 1 0 1 0
	$[-M_B]_{доп}$	1 0 1 1 0 1 0 0 0	
	R_5	0 0 0 0 1 1 0 0 0	0 0 1 0 1 0 1 1
6	$\leftarrow R_5$	0 0 0 1 1 0 0 0 0	0 1 0 1 0 1 1 0
	$[-M_B]_{доп}$	1 0 1 1 0 1 0 0 0	
	R_6	1 1 0 0 1 1 0 0 0	0 1 0 1 0 1 1 0
7	$\leftarrow R_6$	1 0 0 1 1 0 0 0 0	1 0 1 0 1 1 0 0
	$[M_B]_{np}$	0 1 0 0 1 1 0 0 0	
	R_7	1 1 1 0 0 1 0 0 0	1 0 1 0 1 1 0 0
	$M_C \rightarrow$		0 1 0 1 0 1 1 0 0

$$C = (0,101011)_2 \cdot 2^7 = 86.$$

Определим абсолютную и относительную погрешности результата:

$$\Delta C = 86,48648649 - 86 = 0,48648649$$

$$\delta C = \left| \frac{0,48648649}{86,48648649} \right| \cdot 100\% = 0,5625\%$$

Погрешности результатов вызваны неточным представлением операндов. В формате Ф2 операнды представлены точнее и погрешность меньше.